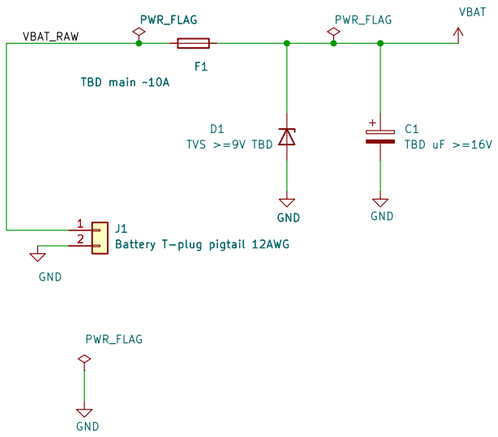
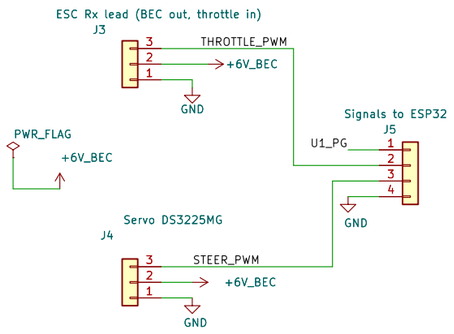
A Battery input (2S LiPo, 6.0–8.4 V)



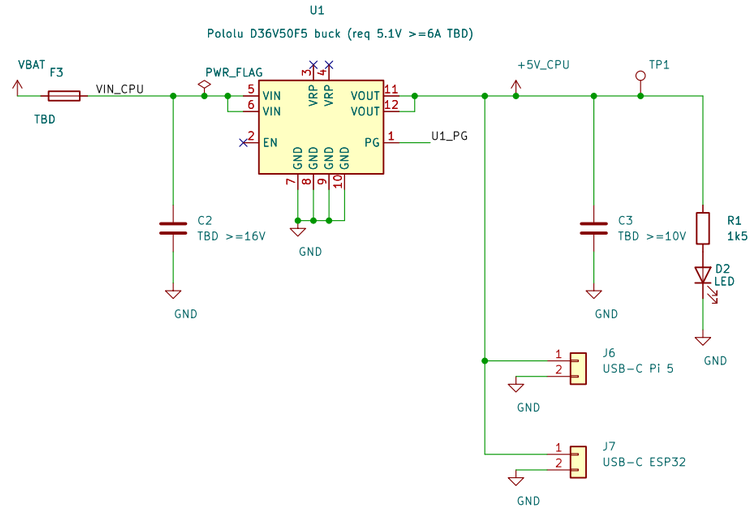
ESC power does NOT pass through this board:
battery -> ESC on its own 12 AWG leads.
J1/F1 carry only the board loads
(about 7.3 A worst case at 6.4 V).

B ESC signals and servo

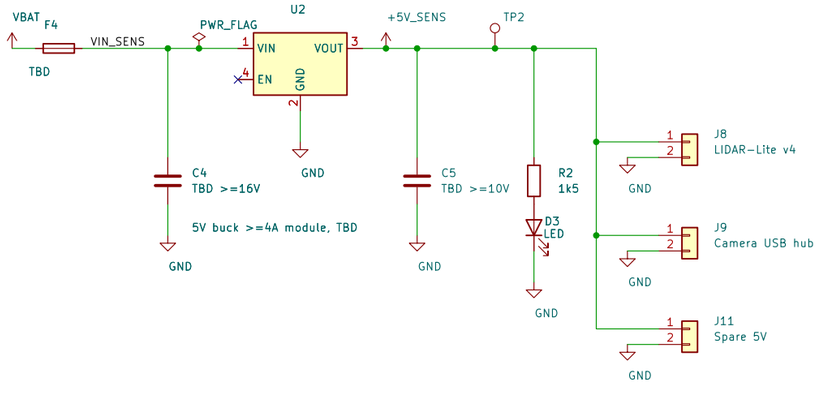


ESC: up to 20 A motor current (THW-1060: 60 A cont.)
from the battery on its own 12 AWG leads, OFF-BOARD.
Only its 3-wire receiver lead lands here (J3).
J5 to ESP32: 1 U1_PG (GPIO TBD), 2 THROTTLE_PWM
(GPIO5), 3 STEER_PWM (GPIO4), 4 GND. 3.3 V logic.

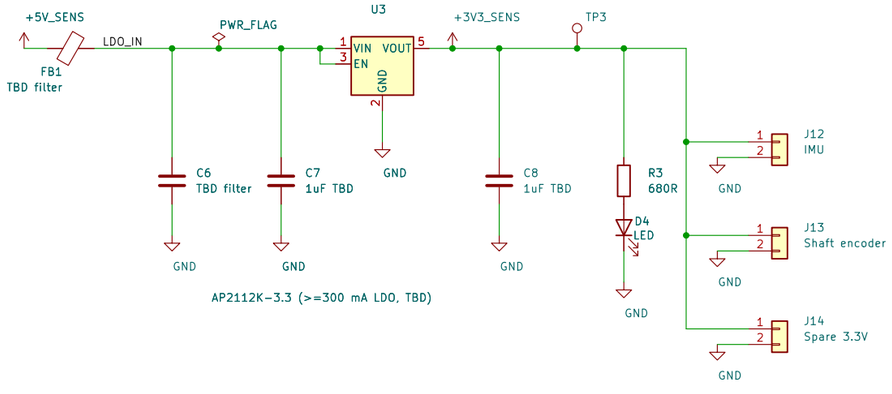
C Compute 5 V (+5V_CPU: Pi 5 + ESP32 only)



D Sensor 5 V (+5V_SENS: heavy sensors + 3.3 V LDO)



E Sensor 3.3 V (+3V3_SENS: LDO from +5V_SENS)



ESP32 3V3 pin does not power these sensors (ESP32 powers nothing).

Notes

1. Battery: 2S LiPo, 6.0–8.4 V (ESC LVC cuts at 6.0 V). T-plug pigtail (12 AWG bare ends) soldered to J1 pads; pad size TBD (footprint is the 2.5 mm2 placeholder).
2. ESC is powered OFF-BOARD: battery -> ESC on its own 12 AWG leads (up to 20 A). This board carries only its own loads (about 7.3 A worst case), so F1 is sized for that, not for the motor.
3. All fuses and protection values TBD.
4. Single common GND; star point at J1 in layout.
5. +6V_BEC is ESC-supplied (servo only), never joined to a 5 V rail.
6. Signals are 3.3 V logic or lower.
7. U1 Pololu D36V50F5: all 12 pins drawn (2x VIN, 2x VOUT, 4x GND, 2x VRP, EN, PG). VRP unused (sensor buck fed from VBAT via F4). EN left unconnected: the module's internal pull-up keeps it on (ESP32 EN control archived). PG: open-drain, to ESP32 via J5 (ESP32 INPUT_PULLUP, non-strapping GPIO TBD).
8. U2 EN per module datasheet (TBD): U2 footprint TBD.
9. Rails are project power symbols (rc_power lib): pin name = Value = net name.
10. PWR_FLAG = ERC marker only (no part, not on the PCB): tells ERC where power enters VBAT_RAW, GND, VBAT, VIN_CPU, VIN_SENS, LDO_IN, +6V_BEC.
11. Net classes (kicad_pro): HighCurrent = *VBAT*, GND; Power5A = +5V_CPU, *VIN_CPU; Power3A = +5V_SENS, *VIN_SENS, *+6V_BEC.
12. J6/J7: 2-wire USB-C pigtails (20 AWG, 30 cm, about 0.10 V drop at 5 A; trim to 10–15 cm) into the Pi 5 and ESP32 USB-C inputs, keeping their own input protection. Pi 5 gets no USB-PD 5 A offer (USB ports limited to 600 mA unless set in config.txt).



Summary: docs/schematics/design_overview.md; project KiCad version 10.0
Spec: docs/schematics/power_board_plan.md
Draft for review, values marked TBD are unconfirmed
gvo27/racecar

Sheet: /
File: rc_power_board.kicad_sch

Title: RC Car Power Distribution Board

Size: A3	Date: 2026-10-29	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 1/1